Course structure of B.Tech (Civil Engineering) for batch 2024-28

Semester 3	Pro		Civic and Social Service Internship	0	0	0	0	1
	BSC		Mathematics – 3	3	1	0	4	4
	ESC		Introduction to Artificial Intelligence	3	0	0	3	3
	PC		Building Materials and Construction Technology	3	0	0	3	3
	PC		Building Planning and Computer Aided drawing	1	0	2	3	2
	PC		Solid Mechanics	4	0	0	4	4
	PC		Concrete Technology	2	0	0	2	2
	PC		Material Testing Laboratory	0	0	2	2	1
				16	**1**	**4**	**23**	**20**
Semester 4	ESC		Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC		Industry 4.0 - Lab (Department Specific)	0	0	2	2	1
	OE		Open Elective 1 (From Other School)	3	0	0	3	3
	PC		Fluid Mechanics	3	0	2	5	4
	PC		Engineering Geology and Soil Mechanics	3	0	2	5	4
	PC		Surveying Practices	3	0	2	5	4
	PC		Structural Analysis	3	1	0	4	4
	Pro		**Industrial Orientation**					0
				17	**1**	**8**	**26**	**22**

Teaching Scheme					Mathematics – III Examination Scheme					
L	T	P	C	Hrs. / Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. The primary objective of this course is to provide students with a comprehensive understanding of advanced mathematical transforms and their applications in engineering.
2. To use the concept of Laplace transform and its significance in solving differential equations.
3. Understand the fundamental concepts of difference equation and Z-transform.
4. To use this basic course in upcoming courses in respective specializations in higher classes.

UNIT 1 LAPLACE TRANSFORM **10 Hrs.**

Piecewise continuous functions and exponential functions, Definition, Existence and Properties of Laplace transforms, Heaviside function, Impulse function, Inverse Laplace transform, Properties of inverse Laplace transforms, Convolution theorem, Applications of Laplace Transforms in solving differential equations.

UNIT 2 DIFFERENCE EQUATION AND Z-TRANSFORM **11 Hrs.**

First and second order difference equation with constant coefficient, Solution of difference equation-complementary function-Particular integral, Z - transform, Properties of Z-transforms, Convolution of two sequences, inverse Z-transform, Solution of Difference equations using Z-transform.

UNIT 3 FOURIER TRANSFORM **10 Hrs.**

Fourier Integral, Fourier Cosine and Sine transform, Discrete Fourier Transform, Continuous Fourier Transform, Linearity, time shifting, frequency shifting, and convolution properties, Applications in image processing, communication systems, and quantum mechanics.

UNIT 4 PARTIAL DIFFERENTIAL EQUATIONS **11 Hrs.**

Formation of Partial Differential Equations (PDEs), Solutions of PDEs of the first order, Cauchy problem for first-order PDEs, Lagrange's method, Charpit and Jacobi methods for solving first-order nonlinear PDEs. Classification of second order PDEs, one-dimensional heat equation, wave equation, and Laplace equation, Applications of PDE's to wave analysis.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES:

CO-1: Define the partial differential equations of first and second order in order to model or understand the Engineering applications.

CO-2: Understand the techniques of Laplace transforms, Z-transforms, and Fourier transforms to solve critical mathematical problems.

CO-3: Apply the methods of Transform techniques to solve differential equations involving piecewise continuous and exponential functions.

CO-4: Classify the partial differential equations of different orders and solve those using different techniques.

CO-5: Appraise engineering or scientific problems that can be addressed using transform techniques.

CO-6: Formulate and create various application problems in engineering using series and Transform techniques.

TEXT/REFERENCE BOOKS:

1. R.K. Jain & S.R.K. Iyengar, **"Advanced Engineering Mathematics"**, Narosa Publishing House.
2. E. Kreyszig,**"Advanced Engineering Mathematics"**, John Wiley & Sons.
3. Peter V. O'Neil**, "Advanced Engineering Mathematics"**, Cengage Learning.
4. K. Sankara Rao, **"Introduction to Partial Differential Equations"**, PHI Learning.
5. T. Amaranth,**" An Elementary Course in Partial Differential Equations"**, Narosa Publishing House, New Delhi.
6. Glynn James, **"Advanced Modern Engineering Mathematics"**, Pearson.

					Introduction to Artificial Intelligence					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	03	25	50	25	--	--	100

COURSE OBJECTIVES

1. Understand key concepts and applications of Artificial Intelligence (AI).
2. Master statistical analysis and data visualization with Python.
3. Gain proficiency in implementing various machine learning algorithms.
4. Develop practical skills in data preparation, model training, and evaluation.
5. Apply regression and classification algorithms to solve real-world problems.

Unit-1: INTRODUCTION ARTIFICIAL INTELLIGENCE **08 Hours.**

Introduction to Artificial Intelligence; Applications and Techniques of AI; Applications in Engineering: Predictive maintenance, quality control, design optimization, autonomous systems; Ethical and Social Implications: Bias, privacy, job displacement; Future Trends: Explainable AI, Ethics, Regulation; Future of Work; Real-world AI applications; AI Career Advice

UNIT-2: STATISTICS AND DATA VISUALIZATION **10 Hours.**

Introduction to Statistics and Data Visualization; Descriptive Statistics (Mean, Median, variance etc) with NumPy and Pandas; Visualization (Line Plot, Scatter Plot, Bar Chart, Histogram, Pie Chart, Box Plot, Heatmap, Area Plot, Contour Plot); Customizing Plots for Better Presentation; Exploratory Data Analysis (EDA); Real-World Case Studies and Examples, Hands-On Exercises using Python/R etc.

UNIT-3: MACHINE LEARNING ALGORITHMS **12 Hours.**

Introduction to Machine Learning, ML Techniques: Supervised Learning, Unsupervised Learning and Reinforcement Learning; ML applications across different sectors, ML Data Preparation, Training and Test Sets, Regression Models (Linear Regression, Decision Tree, Random Forest, Support Vector etc); Accuracy Assessment; Model Comparison; Practice on real world datasets.

Unit-4: SUPERVISED AND UNSUPERVISED LEARNING **12 Hours.**

Difference between Regression and Classification; Working Principles of Unsupervised and Supervised Classification Algorithms: K-Nearest Neighbor, Linear Regression, Logistic Regression, Support Vector Machine; Clustering Algorithms: Hierarchical, Agglomerative Clustering, Partitional etc. Confusion Matrix, Accuracy Assessment, Hyperparameters; Practice on real world datasets.

Max. 42 Hours.

COURSE OUTCOMES

CO1: Understand the principles and applications of Artificial Intelligence (AI) in various fields.
CO2: Apply statistical analysis and data visualization techniques to explore and present data effectively.
CO3: Demonstrate proficiency in data preparation, model training, and evaluation for machine learning tasks.
CO4: Utilize regression algorithms to solve real-world problems.
CO5: Apply classification algorithms to solve real-world problems.
CO6: Analyze and interpret the results of machine learning models to make data-driven decisions.

TEXT/REFERENCE BOOKS

1. Russell, S., & Norvig, P. (2016). **"Artificial Intelligence: A Modern Approach"**, Pearson.
2. McKinney, W. (2017). **"Python for Data Analysis"**, O'Reilly Media.
3. Murphy, K. P. (2012). **"Machine Learning: A Probabilistic Perspective"**. MIT Press.
4. Provost, F., & Fawcett, T. (2013). **"Data Science for Business"**. O'Reilly Media.
5. Müller, A. C., & Guido, S. (2016). **"Introduction to Machine Learning with Python"**. O'Reilly Media.

Teaching Scheme					Building Materials and Construction Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES:

1. To understand the concept of materials used in building construction.
2. To introduce the different building components.
3. To study the difference between super structure and sub structure used in construction engineering.
4. To study techniques used in construction engineering.

UNIT I: BUILDING MATERIALS AND COMPONENTS **12 Hrs.**

Materials: Natural: Stone, brick, aggregates types and properties. Artificial: cement types and properties, concrete blocks (LWC, AAC) admixtures, supplementary cementitious materials, timber, aluminIum, plastics, Modern materials: Glass, Ceramics, Sealants for joints, Fibre glass reinforced plastic, clay products, composite materials, fibre textiles (Geomembranes and Geotextiles for earth reinforcement), recycled concrete aggregates.

UNIT II: BUILDING COMPONENTS **08 Hrs.**

Foundation- types, walls, beams, columns, arches, trusses, lintels, staircase and escalator, flooring, roof (truss, terrace, prefab roof), Bridges, girders and offshore platforms.

UNIT III: CONSTRUCTION PRACTICES **10 Hrs.**

Substructures: Earthwork, formwork (mivan), dewatering, Superstructures: stone masonry, brick masonry, scaffoldings – erection and dismantling, damp and water proofing, plastering, paint and varnish for wood, waterproofing, termite proofing, and fire proof, thermal and sound insulation.

UNIT IV: CONSTRUCTION TECHNOLOGY AND EQUIPMENT **12 Hrs.**

Construction technology: Shotcrete, techniques of box jacking, pipe jacking- under water construction of diaphragm walls and basement. Tunnelling and piling techniques, Precast and Prefabricated structures, Prestressed concrete and types, Construction Equipment: Earth movers, Excavators, handling and hauling equipment, mixers, compactors, dewatering equipment, equipment for erection of structures, trenching and tunneling.

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES:

CO-1: Identify the latest materials used in buildings their applications
CO-2: Understand different components of buildings and their applications
CO-3: Understand and apply different practices for sub structure
CO-4: Understand and apply different practices for super structure
CO-5: Evaluate different construction technology methods.
CO-6: Examine different construction equipment used in construction

TEXT/REFERENCE BOOKS

1. Varghese.P.C, **"Building Materials"**, PHI Learning Pvt. Ltd, New Delhi, 2015.
2. Rajput. R.K., **"Engineering Materials"**, S. Chand and Company Ltd., 2008.
3. Sankar, S.K. and Saraswati, S., **"Construction Technology"**, Oxford University Press, New Delhi, 2008.
4. S.C.Sharma, **"Construction Equipment & Management"**, Kindle Edition, 2018

Teaching Scheme					Building Planning and Computer Aided Drawing					
					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	2	2	3	25	50	25	--	--	100

COURSE OBJECTIVES

5. To learn about different principles of planning and building bye laws and functions of governing bodies.
6. To study the terminology used for building planning and different aspects of the building with fire protection.
7. To learn about plan, elevation and section of the building for residential and commercial buildings.

1. UNIT I: BUILDING BYLAWS AND FUNCTIONS **02 Hrs.**

Introduction: Building industries, Classification of Buildings, Components of Building, Orientation of Building, Principle of Planning, National Building Code of India – 2005, Objectives and Scope of Building Bye Laws, Function of Different Governing Authority. Signs and Symbols with Abbreviations used in Building Drawing, Different types of Doors and Windows.

UNIT II: FUNCTIONAL PLANNING AND BUILDING TERMINOLOGY **03 Hrs.**

Building Planning: Aspect and Prospect of Building, Types of area, Set-back, FSI, Building Bye Laws of Residential Building, Some of Important Terminology related with Building Planning, Fire Protection Kingpost Truss, Queen post Truss, Stairs Drawing of30 cm Wall Section from Foundation to Slab (GF + FF).

UNIT III: PLANNING OF RESIDENTIAL BUILDING **04 Hrs.**

Building Drawings: Types and Site Selection for Residential Building, Minimum, Dimensions of the Rooms for Residential Building, Planning Process and Requirement of Different Rooms of Building, Line Diagram and Site Plan for residential Building Planning of Single Story Residential Building Planning of Double Story Residential Building.

UNIT IV: PLANNING PUBLIC BUILDING AND PERSPECTIVE DRAWING **05 Hrs.**

Planning of Different Public Building. Perspective Drawing: Necessity, Principles and Characteristics of Perspective Drawing Terminology Related with Perspective Drawing, Classification of Perspective Drawing. Planning of Public Building (School, Bank, Commercial Complex, etc- Two Point Perspective Drawing.

TOTAL HOURS: 14 Hrs.

COURSE OUTCOMES

CO-1: Explain local building bye-laws and provisions of national building code in respect of building and town planning.
CO-2: Draw different abbreviations of buildings.
CO-3: Demonstrate the functional aspects of planning and architecture for buildings.
CO-4: Prepare working drawings, foundation plans and other executable drawings with proper details for residential buildings.
CO-5: Prepare working drawings, foundation plans and other executable drawings with proper details for public buildings.
CO-6: Execute building plans, 3d drawings of buildings using software.

TEXT/REFERENCE BOOKS

1. N. Kumara Swamy, A. Kameswara Rao **"Building Planning And Drawing",** Charotar Publications
2. B. P. Verma, **"Civil Engineering Drawing and House Planning",** Publishers.
3. Gurcharan Singh **"Building Planning Designing and Scheduling".**
4. S. S. Bhavikatti (Author), M. V. Chitawa **"Building Planning and Drawing"**.

					Building Planning and Computer Aided Drawing Lab					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	2	2	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the fundamental principles and concepts of planning and architecture for buildings.
2. To study about different views of layout.
3. To learn the development controls covered by building bye laws and national building code for buildings.

LIST OF EXPERIMENTS

1 List out abbreviations and symbols used in building planning. (2 A-2 size sheet).

2 Draw a layout of different types of doors, windows, trusses and stairs used in building. (4 A-2 separate sheet for each component).

3 Draw typical layout of building principles (1 A-1 size sheet).

4 Draw typical layout of Plan – Section – Elevation for Single storied residential building ((1 A-1 size sheet).

5 Draw typical layout of Plan – Section – Elevation for Double storied residential building ((1 A-1 size sheet).

6 Draw typical layout of Plan – Section – Elevation for Public Building (Bank, School, Commercial Complex) (1 A-1 size sheet).

7 Draw a perspective drawing of building (1 A-1 size sheet).

8 Draw typical layout of Plan – Section – Elevation for Double storied residential building in Auto-CAD with basic commend (1 A-1 size sheet printed).

COURSE OUTCOMES

Co-1: Draw different abbreviations of buildings.

Co-2: Demonstrate the fundamental principles, concepts of planning and architecture for buildings.

Co-3: Prepare working drawings, foundation plans and other executable drawings with proper details for buildings.

Co-4: Demonstrate local building bye-laws and provisions of national building code in respect of building.

Co-5: Explain building bye laws and principles of planning for residential and public buildings.

Co-6: Execute building plans, 3d drawings of buildings using software.

TEXT/REFERENCE BOOKS

1. **NATIONAL BUILDING CODE.**
2. S.C.Rangwala, **"CIVIL ENGINEERING DRAWING".**
3. Dr. H.J. Shah, **"Building Planning and Drawing",** Charter Publishing House Pvt. Limited 2007.

Teaching Scheme					Solid Mechanics					
					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
4	0	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES:

1. To analyze stress and strain in structural components under different loading conditions.
2. To apply theoretical concepts to solve engineering problems related to material behavior.
3. To design and evaluate the structural integrity of components using industry-standard methods

UNIT I: COMPOUND STRESSES, SFD AND BMD **16 Hrs.**

Review of forces, moments, equilibrium, and free-body diagrams. Shear force and bending moment diagrams for statically determinate beams with complex loading. Compound stresses- Analytical and graphical method.

UNIT II: FLEXURAL AND SHEAR STRESSES IN BEAMS **16 Hrs.**

Introduction to CG and MI, Simple bending theory, flexural stresses, shear stresses, shear centre, Uniform torsion.

UNIT III: COMBINED STRESSES **12 Hrs.**

Stresses under the combined action of direct loading and bending moment, determination of stress in the case of retaining walls, chimney and dams, overturning and sliding, stress due to direct loading and bending moment about both axis.

UNIT IV: SLOPE AND DEFLECTION **12 Hrs.**

Deflection of beams: Introduction, Elastic curve derivation of differential equation of flexure, Sign convention, Slope and deflection for standard loading classes using Engineering methods

Standard cases of end conditions and corresponding effective length, buckling, principal axes, Euler's and Rankine's formula

TOTAL HOURS: 56 Hrs.

COURSE OUTCOMES:

CO-1: To introduce students to the principles of statics and dynamics applied to engineering structures.
CO-2: To comprehend the fundamental principles of stress and strain in solid materials.
CO-3: To analyze and calculate the deformations in structural elements subjected to various loading conditions.
CO-4: To analyze and calculate the stresses in structural elements subjected to various loading conditions.
CO-5: To apply theoretical concepts and analytical techniques to solve problems related to the behavior of structural components.
CO-6: To evaluate the mechanical properties of materials commonly used in civil engineering practice.

TEXT/REFERENCE BOOKS

1. Shah, H. J., "**Mechanics of Solids**", Charotar Publishing House.
2. Subramanian, R., "**Strength of Materials**", Oxford University Press.
3. Junnarkar, S. B., & Shah, H. J., **"Mechanics of Structures, Vol. - I & Ii**", Charotar Publishing House.
4. Kassimali, A., "**Structural Analysis**", Cengage Learning India.
5. Hibbeler, R. C., "**Mechanics of Materials**". Pearson.
6. Beer, F. P., Johnston, E. R., & Dewolf, J.T., **"Mechanics of Materials"**. Tata Mcgraw-Hill
7. Popov, E. P., "**Mechanics of Materials**". Pearson.
8. Megson, T.H.G., "**Structural and Stress Analysis**", Butterworth - Heinemann

Concrete Technology										
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
2	0	0	2	2	25	50	25	--	--	100

COURSE OBJECTIVES:

8. To understand the concept of materials used in concrete construction.
9. To introduce the difference between binders, mortars and concrete.
10. To study the mix design of modern concrete and their applications in construction engineering.
11. To study the hardened and durability properties of concrete and their applications to construction engineering.

UNIT I: CONCRETE INGREDIENTS **08 hrs.**

CEMENT: Portland cement – chemical composition – Hydration, Setting of cement – Structure of hydrate cement – Test on physical properties – Different grades of cement

AGGREGATES: Classification of aggregate – Particle shape & texture –, strength & other mechanical properties of aggregate – Specific gravity, Bulk density, porosity, adsorption & moisture content of aggregate – Bulking of sand – Deleterious substance in aggregate – Soundness of aggregate – Alkali aggregate reaction – Thermal properties – Sieve analysis – Fineness modulus – Grading curves – Grading of fine & coarse Aggregates – Gap graded aggregate – Maximum aggregate size.

MINERAL ADMIXTURES: Introduction to supplementary cementing materials, sustainability and pozzolans. Fly ash, blast furnace slag, silica fume, and metakaolin – their production, properties, and effects on concrete properties; applications

CHEMICAL ADMIXTURES: Water reducers, air entrainers, set controllers, and effects on concrete properties.

UNIT II: MIX DESIGN OF CONCRETE **06 hrs.**

Mix Design as per latest IS method; particle packing density, Cement Mortars, Mortar specifications, types and tests of mortars.

Ready Mixed Concrete Design: Mixing, transport, and placement; consolidation, finishing, and curing of concrete; Extreme weather concreting, special concreting methods.

UNIT III: FRESH AND HARDENED PROPERTIES OF CONCRETE **07 hrs.**

Fresh Concrete: Workability – Factors affecting workability – Measurement of workability by different tests – Setting times of concrete – Effect of time and temperature on workability – Segregation & bleeding – Mixing and vibration of concrete – Steps in manufacture of concrete – Quality of mixing water. Hardened Properties: Compressive strength and parameters affecting it; Tensile strength, Flexural Strength and Bond Strength – direct and indirect; Factors affecting parameters.

UNIT IV: DURABILITY PROPERTIES OF CONCRETE **07 hrs.**

Introduction to durability: relation between durability and permeability, Chemical attack of concrete corrosion of steel rebar, other durability issues. Non-destructive testing methods – codal provisions for NDT. Elasticity, Creep & Shrinkage – Modulus of elasticity – Dynamic modulus of elasticity – Poisson's ratio – Creep of concrete – Factors influencing creep – Relation between creep & Shrinkage. QA/QC of concrete.

Total Hours: 28 hrs.

COURSE OUTCOMES:

On completion of the course, student will be able to:

CO-1: Identify the modern concrete ingredient, properties and their uses
CO-2: Understand different binder and mortars ratios and their applications
CO-3: Create mix design of concrete as per latest IS and ACI codal provision
CO-4: Classify the different fresh and hardened properties of concrete
CO-5: Evaluate durability properties of concrete.
CO-6: Examine the quality control concrete

TEXT/REFERENCE BOOKS

1. R. Santhakumar, **"Concrete Technology"** oxford university press, 2011.
2. M.S. Shetty, **"Concrete Technology- Theory and Practice"**, S.Chand Publication.
3. M.L.Gambhir, **"Concrete Technology"**, TaTaMacgrawhill publication
4. A.M.Neville, **"Concrete Technology"**, Pearson education India ltd
5. IS:10262-2019, IS:4031 (Part 1 to 15), IS;383-1970

					Material Testing Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

4. To illustrate different material properties and their applications.
5. To test the performance of different binders, mortars ratios and different graded concrete
6. To simulate hardened and durability properties of concrete.

LIST OF EXPERIMENTS:

1 To study and evaluate the fineness modulus of fine and coarse aggregates.
2 To study flakiness and elongation index test for different graded coarse aggregates.
3 To study specific gravity of fine and coarse aggregates.
4 To Study fineness of cement by sieve analysis.
5 To Study the specific gravity of cement
6 To study consistency test of cement
7 To study initial and final setting properties of cement
8 To study soundness of cement by Le-Chatelier method
9 To study compressive strength of binder and mortar cubes
10 To study fresh properties of concrete by different methods.
11 To measure hardened properties of concrete.
12 To study Non-Destructive test of concrete by Rebound Hammer test.
13 To measure bending stress on wooden beam/concrete beam
14 To Study tensile test of mild steel.

COURSE OUTCOMES

CO-1: Understand engineering properties of aggregates by performing laboratory tests necessary for building construction.
CO-2: evaluate engineering properties of cement by performing laboratory tests necessary for building design and construction.
CO-3: Compute the grade & properties of concrete by performing laboratory tests necessary for building construction
CO-4: Select and design different mortar ratios
CO-5: Create smart concrete mix design which can be used for cost effective construction of building
CO-6: Determine the mechanical and durability test results and evaluate the suitability for construction purpose

TEXT/REFERENCE BOOKS

1. R. Santhakumar, **"Concrete Technology",** oxford university press, 2011.
2. M.S. Shetty, "**Concrete Technology- Theory and Practice"**, S.Chand Publication.
3. M.L.Gambhir, **"Concrete Technology"**, TaTa Macgrawhill publication
6. A.M.Neville, **"Concrete Technology"**, Pearson education India ltd